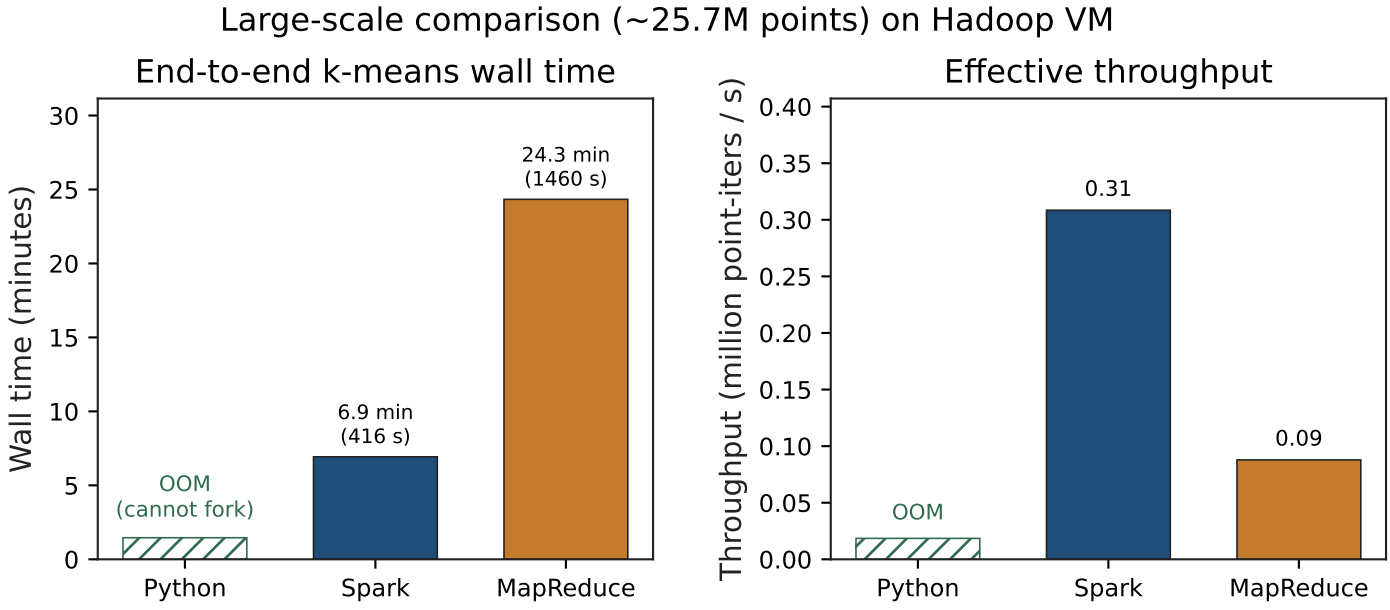

n=25,655,817 (3× full 2018+ scaled), k=4, 5 iters, 4 vCPU / 6 GB.

Python ProcessPool+NumPy OOMs on fork; Spark 1.2.2 (~416 s) beats MapReduce Streaming (~1460 s).